

Spiral Similarities and Linearity

Eric Shen

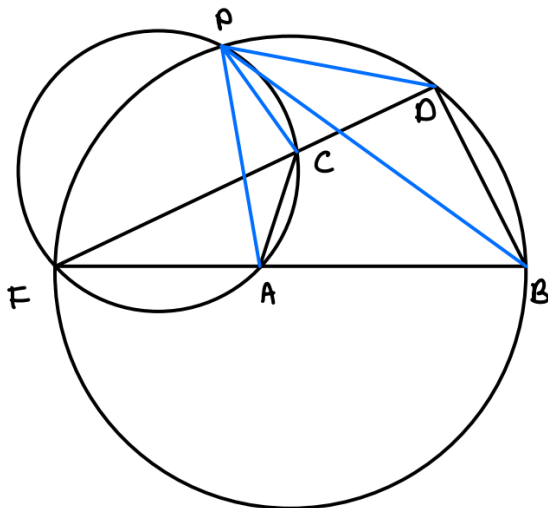
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This is a harder variant of my *Spiral Similarities and Ratios* handout. In that handout, we discussed how spiral similarities interact fruitfully with length ratios, and how we may use such relations to deal with angles and circles which are otherwise difficult to manipulate. In this handout we delve deeper into this relation, discussing how spiral similarities correspond to pairs of linearly moving points, and how we may use this relation to great effect.

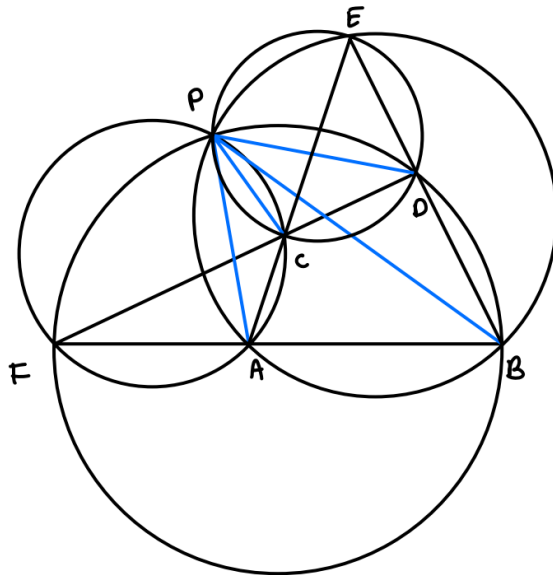
1 Main Lecture

1.1 Reviewing Spiral Similarities

Given any four distinct points A, B, C, D , there exists a **unique** point P such that the triangle PAB is directly similar to PCD . We call P the *spiral center* taking AB to CD . To construct this point, consider $AB \cap CD = F$. Note that $\angle PDC = \angle PBA$ is equivalent to $P \in (FBD)$, and similarly we must also have $P \in (FCA)$. Hence the intersection of these two circles is the desired point P .



By angle chasing and length ratios, we may find that if PAB is directly similar to PCD , then PAC is directly similar to PBD . So we retrieve two pairs of similar triangles.



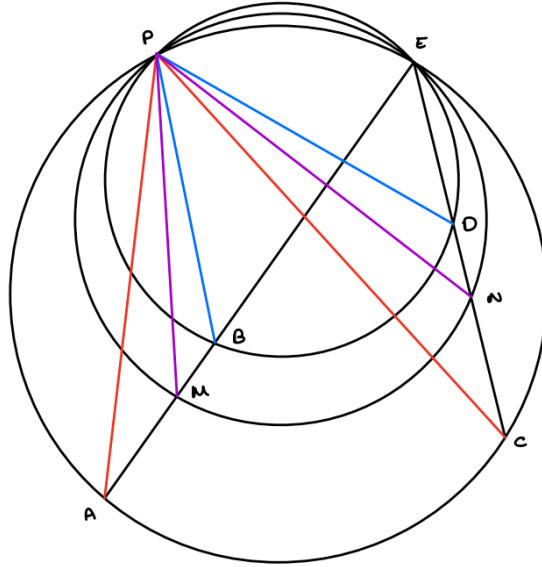
This is useful for two reasons. One, from one pair of similar triangles we have found another that came automatically! Moreover, our construction of P still applies to this other pair: if $AC \cap BD = E$, then $P = (ABE) \cap (CDE)$. So in fact this implies that there are four circles here— (ABE) , (CDE) , (ACF) , and (BDF) —and they all share the common point P , which is also the spiral center. P is hence also known as the *Miquel point* of $ABDC$.

1.1.1 Exercises

1. Let ABC be a triangle, and let $D \in BC$. Let O_1 and O_2 be the circumcenters of ABD and ABC , respectively. Then prove that $AO_1O_2 \sim ADC$.
2. Consider an arbitrary quadrilateral $ABCD$ such that $AB \cap CD = P$, $BC \cap DA = Q$. Let the circumcenters of PBC , PDA , QAB , and QCD be O_1 , O_2 , O_3 , and O_4 , respectively. Prove that $O_1O_2O_3O_4$ is cyclic, and the circumcircle of this cyclic quadrilateral passes through the Miquel point of $ABCD$.
3. Let ABC be a triangle, and let D be a point on side BC . A line through D intersects side AB at X and ray AC at Y . The circumcircle of triangle BXD intersects the circumcircle ω of triangle ABC again at point Z distinct from point B . The lines ZD and ZY intersect ω again at V and W respectively. Prove that $AB = VW$.
4. Circles S_1 and S_2 intersect at points P and Q . Distinct points A_1 and B_1 (not at P or Q) are selected on S_1 . The lines A_1P and B_1P meet S_2 again at A_2 and B_2 respectively, and the lines A_1B_1 and A_2B_2 meet at C . Prove that, as A_1 and B_1 vary, the circumcentres of triangles A_1A_2C all lie on one fixed circle.
5. Circles ω_1 and ω_2 meet at P and Q . Segments AC and BD are chords of ω_1 and ω_2 respectively, such that segment AB and ray CD meet at P . Ray BD and segment AC meet at X . Point Y lies on ω_1 such that $PY \parallel BD$. Point Z lies on ω_2 such that $PZ \parallel AC$. Prove that points Q, X, Y, Z are collinear.

1.2 Ratios

One very useful property about spiral similarities is that if $PAB \sim PCD$, and M and N are points on AB and CD respectively such that $\frac{AM}{MB} = \frac{CN}{ND}$, then the spiral similarity centered at P taking AB to CD also takes M to N . In other words, we also have that $PAM \sim PCN$ and $PMB \sim PND$.



This property is especially useful because it really allows us to get a lot of difficult-to-reach angles from spiral similarities. Midpoints and the like are often very annoying to deal with - with this condition however, this allows us to work with their angles just like usual.

Here's a classic example:

Example 1.1 (USAMO 2006 6) Let $ABCD$ be a quadrilateral, and let E and F be points on sides AD and BC , respectively, such that $\frac{AE}{ED} = \frac{BF}{FC}$. Ray FE meets rays BA and CD at S and T , respectively. Prove that the circumcircles of triangles SAE , SBF , TCF , and TDE pass through a common point.

Walkthrough:

1. What is $(SAE) \cap (SBF)$ called?
2. What is $(TED) \cap (TFC)$ called?
3. Why are these the same point?

In particular, what this demonstrates is that spiral similarities work really well with **length ratios**. In particular, since every pair of directly similar triangles gives you a new similarity, and these pairs can *themselves* be derived from previous ratios, this implies that often chasing ratios that may seem to be useful may work very well in tandem with spiral similarities.

Example 1.2 (Classical) Let ABC be a triangle and let D, E, F be the projections of the incenter of ABC onto BC, AC, AB respectively. Let the circumcircles of ABC and AEF intersect at K . Prove that KD bisects $\angle BKC$.

Walkthrough:

1. In terms of the angle bisector theorem, what would we need to prove in order to show that KD bisects $\angle BKC$?
2. K is a Miquel point. Identify what quadrilateral it is the Miquel point of, and give a few ratios that result from this spiral similarity.
3. Tie the ratios together to prove the claim.

Example 1.3 (ISL 2005 G5) Let $\triangle ABC$ be an acute-angled triangle with $AB \neq AC$. Let H be the orthocenter of triangle ABC , and let M be the midpoint of the side BC . Let D be a point on the side AB and E a point on the side AC such that $AE = AD$ and the points D, H, E are on the same line. Prove that the line HM is perpendicular to the common chord of the circumscribed circles of triangle $\triangle ABC$ and triangle $\triangle ADE$.

Walkthrough:

1. First, some setup. Let $Y = BH \cap AC$ and $Z = CH \cap AB$. Prove that if $K = (AYZ) \cap (ABC)$ and $K \neq A$, then K also lies on HM .
2. Now, it suffices to prove that $AEKD$ is cyclic. Using spiral similarities, prove that this is equivalent to proving that

$$\frac{ZD}{DB} = \frac{YE}{EC}$$

3. Prove this ratio. It may help to recall that $ZC \cap YB = H$.

Note that sometimes it's helpful not just to tie together ratios via spiral similarities, but entire similar figures. For example, if we have the spiral center P taking AB to CD , and E and F are points such that ABE and CDF themselves are directly similar, then this spiral similarity also takes E to F (as the two triangle similarities imply that $PABE \sim PCDF$.)

Example 1.4 (ISL 2012 G3) In an acute triangle ABC the points D, E and F are the feet of the altitudes through A, B and C respectively. The incenters of the triangles AEF and BDF are I_1 and I_2 respectively; the circumcenters of the triangles ACI_1 and BCI_2 are O_1 and O_2 respectively. Prove that I_1I_2 and O_1O_2 are parallel.

Walkthrough:

1. Draw a good diagram :)
2. First off; there's no reason not to add the third incenter I_3 of CDE , so draw that in too. What circles do this incenter lie on (don't prove anything yet)?
3. Let's now prove that AI_1I_2B is cyclic. Prove that this is equivalent to proving that $I_1I_2F \sim ADF$.
4. Let's flip the spiral sim, and prove that $AI_1F \sim DI_2F$. Prove that $AEF \sim DBF$, and demonstrate that this implies $AI_1F \sim DI_2F$.
5. With the key claim proved, let's now move on to the problem. Prove that it suffices to show $CI \perp I_1I_2$, where I is the incenter of the triangle.
6. Prove this by angle chasing or otherwise.
7. Where did we use the similar figures argument?

In a sense, what these examples imply is that a good way of thinking about these spiral similarities is as a sort of **algebraic manipulation**: by manipulating these ratios and spiral similarities, we can "reach" a lot of angles that would be previously unobtainable! In particular, some of the harder examples will demonstrate that this approach of algebraic manipulation to push for the exact similarities you want is exactly what is needed to solve many problems.

1.2.1 Exercises

1. In triangle ABC , points P, Q, R lie on sides BC, CA, AB respectively. Let $\omega_A, \omega_B, \omega_C$ denote the circumcircles of triangles AQR, BRP, CPQ , respectively. Given the fact that segment AP intersects $\omega_A, \omega_B, \omega_C$ again at X, Y, Z , respectively, prove that $YX/XZ = BP/PC$.
2. Let $ABCDE$ be a convex pentagon such that

$$\angle BAC = \angle CAD = \angle DAE \quad \text{and} \quad \angle ABC = \angle ACD = \angle ADE.$$

The diagonals BD and CE meet at P . Prove that the line AP bisects the side CD .

3. Triangle ABC is inscribed in circle Ω . The interior angle bisector of angle A intersects side BC and Ω at D and L (other than A), respectively. Let M be the midpoint of side BC . The circumcircle of triangle ADM intersects sides AB and AC again at Q and P (other than A), respectively. Let N be the midpoint of segment PQ , and let H be the foot of the perpendicular from L to line ND . Prove that line ML is tangent to the circumcircle of triangle HMN .
4. Let $ABCD$ be a fixed convex quadrilateral with $BC = DA$ and BC not parallel with DA . Let two variable points E and F lie on the sides BC and DA , respectively and satisfy $BE = DF$. The lines AC and BD meet at P , the lines BD and EF meet at Q , the lines EF and AC meet at R .
Prove that the circumcircles of the triangles PQR , as E and F vary, have a common point other than P .

1.3 Linearity and the Gliding Principle

Let's now delve into this ratio idea deeper. Its setup can be phrased as follows. Given a triangle ABC and a point $P \in (ABC)$, then for any $E \in AB, F \in AC$ we have that $AEPF$ is cyclic iff

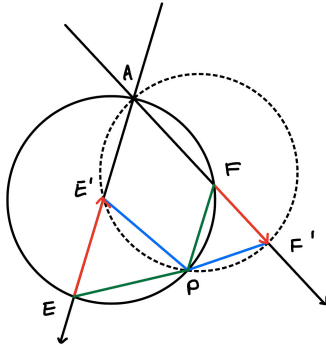
$$\frac{PB}{PC} = \frac{BE}{CF}.$$

Indeed, the given ratio condition is equivalent to PBE being directly similar to PCF , which is equivalent to $AEPF$ being cyclic.

Now, this condition implies that (AEF) passes through a given fixed point iff BE and CF are of a fixed ratio, i.e. for some constants x, y, z , we have that

$$x \cdot AE + y \cdot AF = z.$$

Above, these constants are $x = PC, y = -PB$, and $z = PC \cdot AE - PB \cdot AF$. Observe that here it's important we direct the lengths of AE and AF , so no inconsistencies arise when E or F passes through A .



Let's now flip the setup on its head. If we have E and F moving along two lines which intersect at a unique point A , such that there exist constants x, y, z (not all zero) such that

$$x \cdot AE + y \cdot AF = z,$$

is it true that AEF has a fixed point? Well, in fact we can find this fixed point; take some choice (E', F') of E and F , and pick the (unique) point $P \in (AE'F')$ such that

$$\frac{PE'}{PF'} = -\frac{y}{x}.$$

Here the sign convention implies that P is on the arc $E'F'$ subtended by A if y/x is negative, and on the rest of $(AE'F')$ otherwise. Then by using ratios we can verify that P is indeed fixed as we vary E and F , by the same argument as the above.

Example 1.5 (Classical) Consider a triangle ABC and a point $D \in BC$. Suppose the perpendicular bisectors of BD and DC intersect AB and AC respectively at E and F . Prove that (AEF) passes through a fixed point as D varies.

1. Prove the given problem.
2. By picking two choices of E and F , find the fixed point.

Here's another way in which we can utilize this linearity idea. Before we proceed, let's make sure our terminology is clear.

Definition 1.6 A linearly moving point is a point $p = p(t) \in \mathbb{C}$ which is a function of a variable $t \in \mathbb{R}$ of the form $p = a \cdot t + b$ for some fixed $a, b \in \mathbb{C}$.

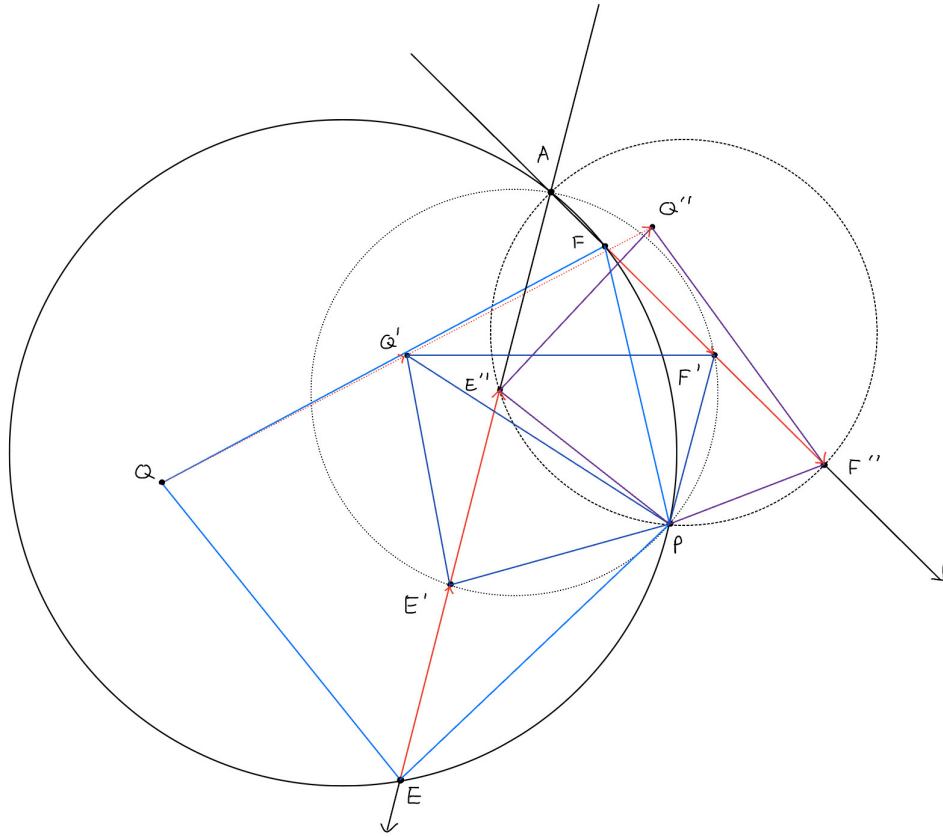
Now, suppose we have linearly moving points $E = e = at + b$ and $F = f = ct + d$ in the time variable $t \in \mathbb{R}$ for fixed $a, b, c, d \in \mathbb{C}$. By the above, there exists a fixed point P such that all triangles PEF are directly similar to each other for all t . Consider another point Q moving alongside them, such that as Q, E, F all move, the triangles QEF made are all similar to each other. This is equivalent to there existing a fixed $k \in \mathbb{C}$ such that

$$\frac{q - e}{f - e} = k \quad \implies \quad q = kf + (1 - k)e.$$

Then this gives that

$$q = k(at + b) + (1 - k)(ct + d) = (ak + c - ck)t + (bk + d - dk).$$

Notably this implies that Q also moves linearly with E and F . So all copies of $PQEF$ are directly similar to each other (which we already knew), but Q also lies on a fixed line! This second fact is also known as the *gliding principle*, and can be very useful for pinning down points moving in spiral similarities.



Here's the canonical example.

Example 1.7 (ISL 2016 G5) Let D be the foot of perpendicular from A to the Euler line (the line passing through the circumcentre and the orthocentre) of an acute scalene triangle ABC . A circle ω with centre S passes through A and D , and it intersects sides AB and AC at X and Y respectively. Let P be the foot of altitude from A to BC , and let M be the midpoint of BC . Prove that the circumcentre of triangle $XS Y$ is equidistant from P and M .

1. Move X linearly, i.e. express it as $x = at + b$ for some varying $t \in \mathbb{R}$ and $a, b \in \mathbb{C}$. Prove that Y , hence S , also move linearly.
2. Let the circumcenter of $XS Y$ be T . Prove T moves linearly, and thus lies on a fixed line.
3. By the above, it suffices to check the given result by checking that two possibilities for T lie on the given line (i.e. the perpendicular bisector of PM). Pick two convenient choices for X (really ω) to check this.

Let's end on a difficult example exhibiting the power of linearity.

Example 1.8 (ISL 2012 G6) Let ABC be a triangle with circumcenter O and incenter I . The points D, E and F on the sides BC, CA and AB respectively are such that $BD + BF = CA$ and $CD + CE = AB$. The circumcircles of the triangles BFD and CDE intersect at $P \neq D$. Prove that $OP = OI$.

1. First, pick a familiar choice for D, E , and F . In this case what is P , and why is the given statement true? We'll call this choice of P K from now on, and we'll call the choices of D, E, F as A_1, B_1, C_1 respectively.
2. Now for an arbitrary choice of D , show that (AEF) passes through a fixed point A_0 .

3. Analogously say (BDF) passes through a fixed point B_0 and (CDE) a fixed point C_0 . Prove that A_0 is on AI (and similarly for the other two points).
4. Prove that A_0, B_0, C_0 lie on the circle with diameter IK .
5. Prove that for a generic choice of D , P also lies on the circle with diameter IK , and finish from here.

1.4 Linearity Exercises

Here are some targeted exercises which focus on these linearity ideas.

1. In $\triangle ABC$, a point D lies on line BC . The circumcircle of ABD meets AC at F (other than A), and the circumcircle of ADC meets AB at E (other than A). Prove that as D varies, the circumcircle of AEF always passes through a fixed point other than A , and that this point lies on the median from A to BC .
2. Let ABC be a triangle with altitudes BE and CF . Let X and Y be the midpoints of BF and CE , respectively, and consider a point Z on the same side of BC as A such that XYZ and ABC are similar but oppositely oriented. Then prove $BZ = CZ$.
3. The angles B and C of an acute-angled triangle ABC are greater than 60° . Points P, Q are chosen on the sides AB, AC respectively so that the points A, P, Q are concyclic with the orthocenter H of the triangle ABC . Point K is the midpoint of PQ . Prove that $\angle BKC > 90^\circ$.

1.5 Review

1. The **Miquel point** P of a quadrilateral $ABCD$ is the spiral center taking AB to DC and AD to BC . If $AB \cap CD = E$ and $AD \cap BC = F$, then P lies on (ADE) , (BCE) , (ABF) , and (CDF) .
2. If $M, N \in AB, CD$ respectively such that $\frac{AM}{MB} = \frac{DN}{NC}$, then we have that P also takes AMB to DMC . This property allows us to deal with the angles of points that aren't as tied-into the diagram as the other points.
 - (a) The above also applies for points M, N satisfying that $AMB \sim DMC$.
3. Any two points E and F linearly moving along lines l and l' , satisfy that if $l \cap l' = A$, then (AEF) passes through a fixed point. We can use this fact to retrieve fixed points out of linear movement in general.
4. Given two linearly moving points E and F , and another varying point P such that all triangles PEF are similar to each other, we have that P is also linearly moving. Notably, P lies on a fixed line! This is known as the *Gliding principle*.

2 Problems

The exercises below are focused mostly on the “ratios” ideas. I’ve split the problems into three parts: Part A (IMO 1 or easier), Part B (IMO 2), and Part C (IMO 3). ★s represent problems I think are particularly good illustrations of the themes of this lecture.

2.1 Part A

1. Let $\triangle ABC$ be an acute-angled triangle, and let D be the foot of the altitude from C . The angle bisector of $\angle ABC$ intersects CD at E and meets the circumcircle ω of triangle $\triangle ADE$ again at F . If $\angle ADF = 45^\circ$, show that CF is tangent to ω .
2. ★ Let $ABCD$ be a cyclic quadrilateral. Let P, Q, R be the feet of the perpendiculars from D to the lines BC, CA, AB , respectively. Show that $PQ = QR$ if and only if the bisectors of $\angle ABC$ and $\angle ADC$ are concurrent with AC .
3. Let ABC be a triangle. Construct three equilateral triangles outside of ABC , with AB, BC , and CA being one of the sides of each of these equilateral triangles. Prove that the centers of these three equilateral triangles themselves form an equilateral triangle.
4. Let ABC be a fixed acute triangle inscribed in a circle ω with center O . A variable point X is chosen on minor arc AB of ω , and segments CX and AB meet at D . Denote by O_1 and O_2 the circumcenters of triangles ADX and BDX , respectively. Determine all points X for which the area of triangle OO_1O_2 is minimized.
5. Let ABC be a triangle with $AB < AC$. Let D be the intersection point of the internal bisector of angle BAC and the circumcircle of ABC . Let Z be the intersection point of the perpendicular bisector of AC with the external bisector of angle $\angle BAC$. Prove that the midpoint of the segment AB lies on the circumcircle of triangle ADZ .
6. Let ABC be a triangle such that $AB > BC$ and let D be a variable point on the line segment BC . Let E be the point on the circumcircle of triangle ABC , lying on the opposite side of BC from A such that $\angle BAE = \angle DAC$. Let I be the incenter of triangle ABD and let J be the incenter of triangle ACE . Prove that the line IJ passes through a fixed point, that is independent of D .
7. ★ Let $ABCD$ be a parallelogram. Points P and Q lie inside $ABCD$ such that $\triangle ABP$ and $\triangle BCQ$ are equilateral. Prove that the intersection of the line through P perpendicular to PD and the line through Q perpendicular to DQ lies on the altitude from B in $\triangle ABC$.

2.2 Part B

1. ★ The isosceles triangle $\triangle ABC$, with $AB = AC$, is inscribed in the circle ω . Let P be a variable point on the arc $\widehat{BC}$ that does not contain A , and let I_B and I_C denote the incenters of triangles $\triangle ABP$ and $\triangle ACP$, respectively.
Prove that as P varies, the circumcircle of triangle $\triangle PI_BI_C$ passes through a fixed point.
2. Let ABC be a triangle with $\angle C = 90^\circ$, and let H be the foot of the altitude from C . A point D is chosen inside the triangle CBH so that CH bisects AD . Let P be the intersection point of the lines BD and CH . Let ω be the semicircle with diameter BD that meets the segment CB at an interior point. A line through P is tangent to ω at Q . Prove that the lines CQ and AD meet on ω .
3. ★ Let O be the circumcenter of an acute triangle ABC . Line OA intersects the altitudes of ABC through B and C at P and Q , respectively. The altitudes meet at H . Prove that the circumcenter of triangle PQH lies on a median of triangle ABC .
4. Let $ABCD$ be an isosceles trapezoid with $AB \parallel CD$. Let E be the midpoint of AC . Denote by ω and Ω the circumcircles of the triangles ABE and CDE , respectively. Let P be the crossing point of the tangent to ω at A with the tangent to Ω at D . Prove that PE is tangent to Ω .

5. Let ABC be an acute-angled triangle with circumcenter O . Let I be a circle with center on the altitude from A in ABC , passing through vertex A and points P and Q on sides AB and AC . Assume that

$$BP \cdot CQ = AP \cdot AQ.$$

Prove that I is tangent to the circumcircle of triangle BOC .

6. ★ Points A_1, B_1, C_1 are chosen on the sides BC, CA, AB of a triangle ABC respectively. The circumcircles of triangles $AB_1C_1, BC_1A_1, CA_1B_1$ intersect the circumcircle of triangle ABC again at points A_2, B_2, C_2 respectively ($A_2 \neq A, B_2 \neq B, C_2 \neq C$). Points A_3, B_3, C_3 are symmetric to A_1, B_1, C_1 with respect to the midpoints of the sides BC, CA, AB respectively. Prove that the triangles $A_2B_2C_2$ and $A_3B_3C_3$ are similar.
7. Let ABC be a triangle and let M and N denote the midpoints of $\overline{AB}$ and $\overline{AC}$, respectively. Let X be a point such that $\overline{AX}$ is tangent to the circumcircle of triangle ABC . Denote by ω_B the circle through M and B tangent to $\overline{MX}$, and by ω_C the circle through N and C tangent to $\overline{NX}$. Show that ω_B and ω_C intersect on line BC .
8. Let $ABCD$ be a quadrilateral with $AC = BD$. Diagonals AC and BD meet at P . Let ω_1 and O_1 denote the circumcircle and the circumcenter of triangle ABP . Let ω_2 and O_2 denote the circumcircle and circumcenter of triangle CDP . Segment BC meets ω_1 and ω_2 again at S and T (other than B and C), respectively. Let M and N be the midpoints of minor arcs $\widehat{SP}$ (not including B) and $\widehat{TP}$ (not including C). Prove that $MN \parallel O_1O_2$.
9. Let ABC be a triangle with circumcircle Γ and $AB \neq AC$. Let D and E lie on the arc BC of Γ not containing A such that $\angle BAE = \angle DAC$. Let the incenters of BAE and CAD be X and Y , respectively, and let the external tangents of the incircles of BAE and CAD intersect at Z . Prove that Z lies on the common chord of Γ and the circumcircle of AXY .
10. ★ Given a triangle ABC with $AB > BC$, let Ω be the circumcircle. Let M, N lie on the sides AB, BC respectively, such that $AM = CN$. Let K be the intersection of MN and AC . Let P be the incentre of the triangle AMK and Q be the K -excentre of the triangle CNK . If R is midpoint of the arc ABC of Ω then prove that $RP = RQ$.
11. ★ A convex quadrilateral $ABCD$ has perpendicular diagonals. The perpendicular bisectors of the sides AB and CD meet at a unique point P inside $ABCD$. Prove that the quadrilateral $ABCD$ is cyclic if and only if triangles ABP and CDP have equal areas. (solve this synthetically!)
12. Let ABC be an acute-angled triangle with circumcircle ω and circumcentre O . Points $D \neq B$ and $E \neq C$ lie on ω such that $BD \perp AC$ and $CE \perp AB$. Let CO meet AB at X , and BO meet AC at Y . Prove that the circumcircles of triangles BXD and CYE have an intersection lie on line AO .
13. ★ Let ABC be a triangle and let B_1 and C_1 be variable points on sides $\overline{BA}$ and $\overline{CA}$, respectively, such that $BB_1 = CC_1$. Let $B_2 \neq B_1$ denote the point on $\odot(ACB_1)$ such that BC_1 is parallel to B_1B_2 , and let $C_2 \neq C_1$ denote the point on $\odot(ABC_1)$ such that CB_1 is parallel to C_1C_2 . Prove that as B_1, C_1 vary, the circumcircle of $\triangle AB_2C_2$ passes through a fixed point, other than A .

2.3 Part C

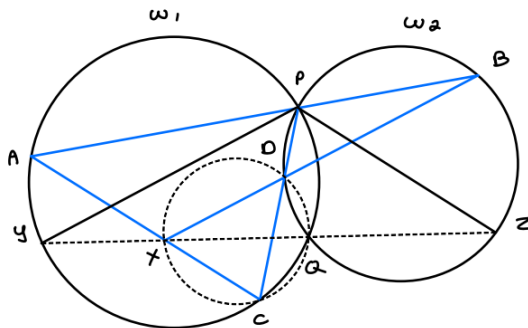
1. Let P be a point on the circumcircle of acute triangle ABC . Let D, E, F be the reflections of P in the A -midline, B -midline, and C -midline. Let ω be the circumcircle of the triangle formed by the perpendicular bisectors of AD, BE, CF . Show that the circumcircles of $\triangle ADP, \triangle BEP, \triangle CFP$, and ω share a common point.
2. Two circles ω_1 and ω_2 intersect at points A and B . Line l is tangent to ω_1 at P and to ω_2 at Q so that A is closer to l than B . Let X and Y be points on major arcs PA (on ω_1) and AQ (on ω_2), respectively, such that $AX/PX = AY/QY = c$. Extend segments PA and QA through A to R and S , respectively, such that $AR = AS = c \cdot PQ$. Given that the circumcenter of triangle ARS lies on line XY , prove that $\angle XPA = \angle AQY$.

3. Let ABC be a fixed acute-angled triangle. Consider some points E and F lying on the sides AC and AB , respectively, and let M be the midpoint of EF . Let the perpendicular bisector of EF intersect the line BC at K , and let the perpendicular bisector of MK intersect the lines AC and AB at S and T , respectively. We call the pair (E, F) *interesting*, if the quadrilateral $KSAT$ is cyclic. Suppose that the pairs (E_1, F_1) and (E_2, F_2) are interesting. Prove that $\frac{E_1E_2}{AB} = \frac{F_1F_2}{AC}$.
4. ★ Let acute triangle ABC have orthocenter H . Let B_1, C_1, B_2 , and C_2 be collinear points which lie on lines AB, AC, BH , and CH , respectively. Let ω_B and ω_C be the circumcircles of triangles BB_1B_2 and CC_1C_2 , respectively. Prove that the radical axis of ω_B and ω_C intersects the line through their centers on the nine-point circle of triangle ABC .
5. Let the sides AD and BC of the quadrilateral $ABCD$ (such that AB is not parallel to CD) intersect at point P . Points O_1 and O_2 are circumcenters and points H_1 and H_2 are orthocenters of triangles ABP and CDP , respectively. Denote the midpoints of segments O_1H_1 and O_2H_2 by E_1 and E_2 , respectively. Prove that the perpendicular from E_1 on CD , the perpendicular from E_2 on AB and the lines H_1H_2 are concurrent.
6. ★ An equilateral pentagon $AMNPQ$ is inscribed in triangle ABC such that $M \in \overline{AB}$, $Q \in \overline{AC}$, and $N, P \in \overline{BC}$. Let S be the intersection of $\overleftrightarrow{MN}$ and $\overleftrightarrow{PQ}$. Denote by ℓ the angle bisector of $\angle MSQ$.
Prove that $\overline{OI}$ is parallel to ℓ , where O is the circumcenter of triangle ABC , and I is the incenter of triangle ABC .
7. ★ In triangle ABC let O and H be the circumcenter and the orthocenter. The point P is the reflection of A with respect to OH . Assume that P is not on the same side of BC as A . Points E, F lie on AB, AC respectively such that $BE = PC, CF = PB$. Let K be the intersection point of AP, OH . Prove that $\angle EKF = 90^\circ$.
8. ★ Let $ABCD$ be a non-cyclical, convex quadrilateral, with no parallel sides. The lines AB and CD meet in E . Let $M \neq E$ be the intersection of circumcircles of ADE and BCE . The internal angle bisectors of $ABCD$ form an convex, cyclical quadrilateral with circumcenter I . The external angle bisectors of $ABCD$ form an convex, cyclical quadrilateral with circumcenter J .
Show that I, J, M are collinear.
9. ★ Let ABC be an acute triangle with circumcenter O and orthocenter H . Let Γ denote the circumcircle of triangle ABC , and N the midpoint of OH . The tangents to Γ at B and C , and the line through H perpendicular to line AN , determine a triangle whose circumcircle we denote by ω_A . Define ω_B and ω_C similarly. Prove that the common chords of ω_A, ω_B and ω_C are concurrent on line OH .

3 Solutions

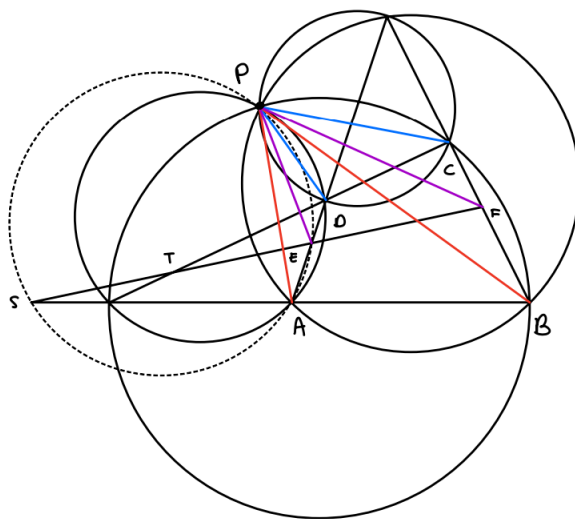
Example 3.1 Circles ω_1 and ω_2 meet at P and Q . Segments AC and BD are chords of ω_1 and ω_2 respectively, such that segment AB and ray CD meet at P . Ray BD and segment AC meet at X . Point Y lies on ω_1 such that $PY \parallel BD$. Point Z lies on ω_2 such that $PZ \parallel AC$. Prove that points Q, X, Y, Z are collinear.

Note that Q is the Miquel point of $APDX$. This implies that $QDXC$ is a cyclic quadrilateral. Now the parallel condition gives us that $\angle DQX = \angle DCX = \angle DPZ = 180^\circ - \angle DQZ$, so X, Q , and Z are collinear. Similarly, since $\angle YQC = \angle YPC = \angle XDC = \angle XQC$, we have that Y, X , and Q are collinear as well, thus solving the problem.



Example 3.2 Let $ABCD$ be a quadrilateral, and let E and F be points on sides AD and BC , respectively, such that $\frac{AE}{ED} = \frac{BF}{FC}$. Ray FE meets rays BA and CD at S and T , respectively. Prove that the circumcircles of triangles SAE , SBF , TCF , and TDE pass through a common point.

Consider the spiral center K taking AE to BF . Since $\frac{AE}{ED} = \frac{BF}{FC}$, we have that this spiral center also takes ED to FC . Hence K is the Miquel point of both $ABFE$ and $EFCD$, and so $K \in (SAE), (SBF), (TED)$, and (TFC) , thus solving the problem.

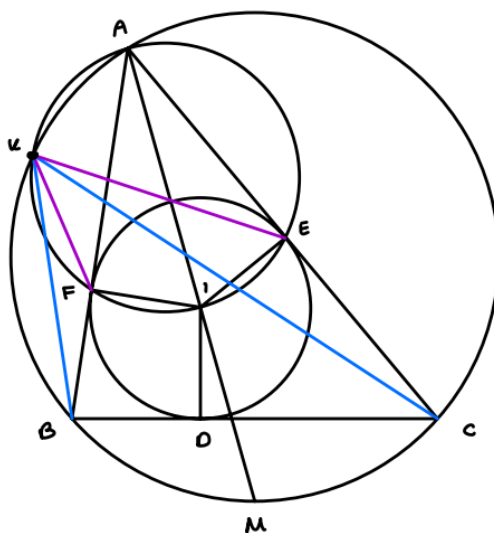


Example 3.3 Let ABC be a triangle and let D, E, F be the projections of the incenter of ABC onto BC, AC, AB respectively. Let the circumcircles of ABC and AEF intersect at K . Prove that KD bisects $\angle BKC$.

Note that K is the spiral center taking FE to BC . This hence implies that:

$$\begin{aligned} \frac{KB}{KC} &= \frac{FB}{EC} \\ &= \frac{DB}{DC} \end{aligned}$$

with the last equality arising from the fact that $FB = DB$ and $EC = DC$ by incircle properties. This thus implies that DK bisects $\angle BKC$, as desired.



Example 3.4 Let $\triangle ABC$ be an acute-angled triangle with $AB \neq AC$. Let H be the orthocenter of triangle ABC , and let M be the midpoint of the side BC . Let D be a point on the side AB and E a point on the side AC such that $AE = AD$ and the points D, H, E are on the same line. Prove that the line HM is perpendicular to the common chord of the circumscribed circles of triangle $\triangle ABC$ and triangle $\triangle ADE$.

Let $Y = BH \cap AC$, $Z = CH \cap AB$, and A' be the antipode of A on (ABC) . note that $BHCA'$ is a parallelogram (both BA' and CH are perpendicular to AB), so A' lies on HM . But now if $K = A'H \cap (ABC)$, then we know that $\angle AKH = \angle AK A' = 90^\circ$. Hence K lies on (AH) , and so $K \in (AYZ)$. Thus, K is the spiral center taking ZY to BC .

Now, note that $\angle ADE = 90^\circ - \frac{1}{2}\angle A = \frac{1}{2}(90^\circ + 90^\circ - \angle A) = \frac{1}{2}(\angle AZC + \angle ABY)$. This thus implies that $\angle DHB = \frac{1}{2}\angle ZHB$, so thus DE bisects $\angle ZHB$. Hence, we have that by the Angle Bisector theorem,

$$\frac{ZD}{DB} = \frac{ZH}{HB}$$

$$\frac{YE}{EC} = \frac{YH}{HC}$$

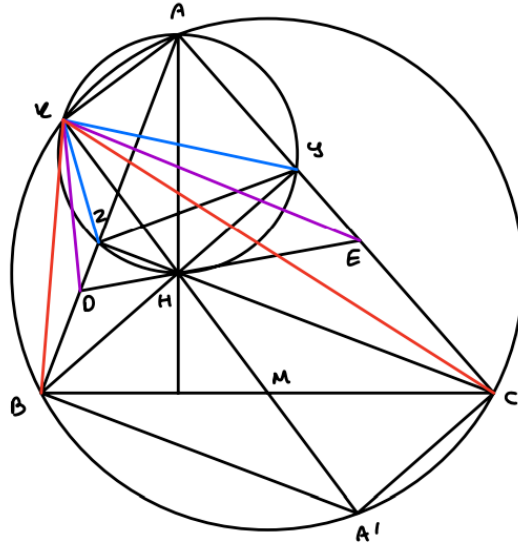
But in fact $\angle HZB = \angle HYC = 90^\circ$, and $\angle H B Z = \angle H C Y$ (as $BZYC$ is cyclic), so thus $HBZ \sim HCY$. This thus implies that:

$$\frac{ZH}{HB} = \frac{YH}{HC}$$

so thus:

$$\frac{ZD}{DB} = \frac{YE}{EC}$$

But now consider K ; we have that since K takes BZ to CY through a spiral sim, it also takes BD to CE . But then K is the Miquel point of $BDEC$, so since $BD \cap CE = A$ we have that $K \in (ADE)$. Since $K \in HM, (ABC)$ as well, this thus solves the problem!



Example 3.5 In an acute triangle ABC the points D, E and F are the feet of the altitudes through A, B and C respectively. The incenters of the triangles AEF and BDF are I_1 and I_2 respectively; the circumcenters of the triangles ACI_1 and BCI_2 are O_1 and O_2 respectively. Prove that I_1I_2 and O_1O_2 are parallel.

The main claim is that AI_1I_2B is cyclic. To prove this, note first that $AEF \sim DBF$ by angle chasing (in particular, this is a spiral similarity). Hence this implies that F takes the figure FAE to FDB , and in this spiral similarity also takes I_1 to I_2 . So $FI_1I_2 \sim FAD$. Now this implies that:

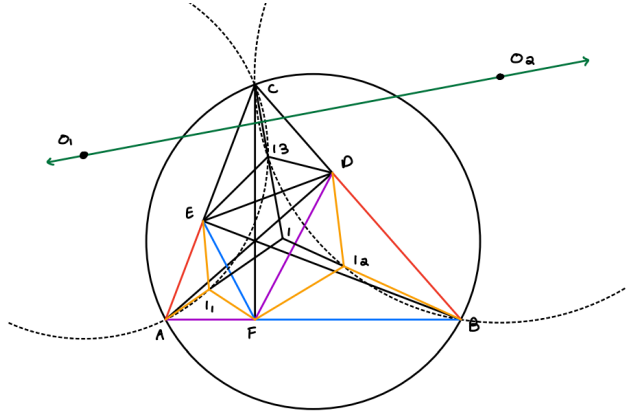
$$\begin{aligned} \angle I_1AB + \angle I_1I_2B &= \frac{1}{2} \angle A + \angle ADF + \angle FI_2B \\ &= \frac{1}{2} \angle A + \angle ACF + 90^\circ + \frac{1}{2} \angle FDB \\ &= \frac{1}{2} \angle A + 90^\circ - \angle A + 90^\circ + \frac{1}{2} \angle A \\ &= 180^\circ \end{aligned}$$

which gives us that AI_1I_2B is cyclic, as desired. ■

Now, let I_3 be the incenter of CDE , and I the incenter of ABC . Note that AI_1, BI_2 , and CI_3 all pass through CI . In addition, by an analogous proof to the one above, we have that BI_2I_3C and CI_3I_1A are both cyclic as well. this implies that $O_1O_2 \perp CI_3$. However, note that:

$$\begin{aligned} \angle CIA &= 90^\circ + \frac{1}{2} \angle B \\ &= 90^\circ + \angle IBA \\ &= 90^\circ + \angle I I_1 I_2 \end{aligned}$$

This thus implies that $I_1I_2 \perp CI$ as well, and hence $I_1I_2 \parallel O_1O_2$, as desired!



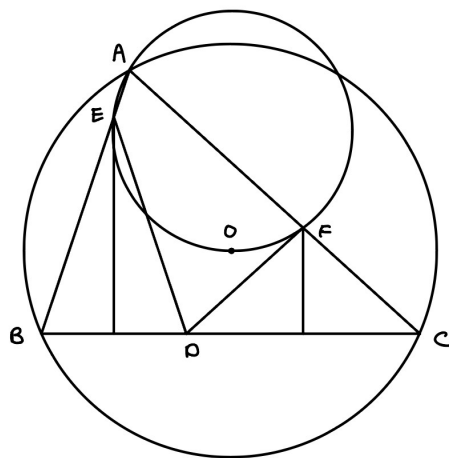
Example 3.6 (Classical) Consider a triangle ABC and a point $D \in BC$. Suppose the perpendicular bisectors of BD and DC intersect AB and AC respectively at E and F . Prove that (AEF) passes through a fixed point as D varies.

We have that

$$BE \sin B + CF \sin C = \frac{1}{2}(BD + DC) = \frac{BC}{2}.$$

This readily implies that there exist x, y, z independent of D , such that we always have $x \cdot AE + y \cdot AF = z$. Hence by our results on linearity we have that (AEF) passes through a fixed point.

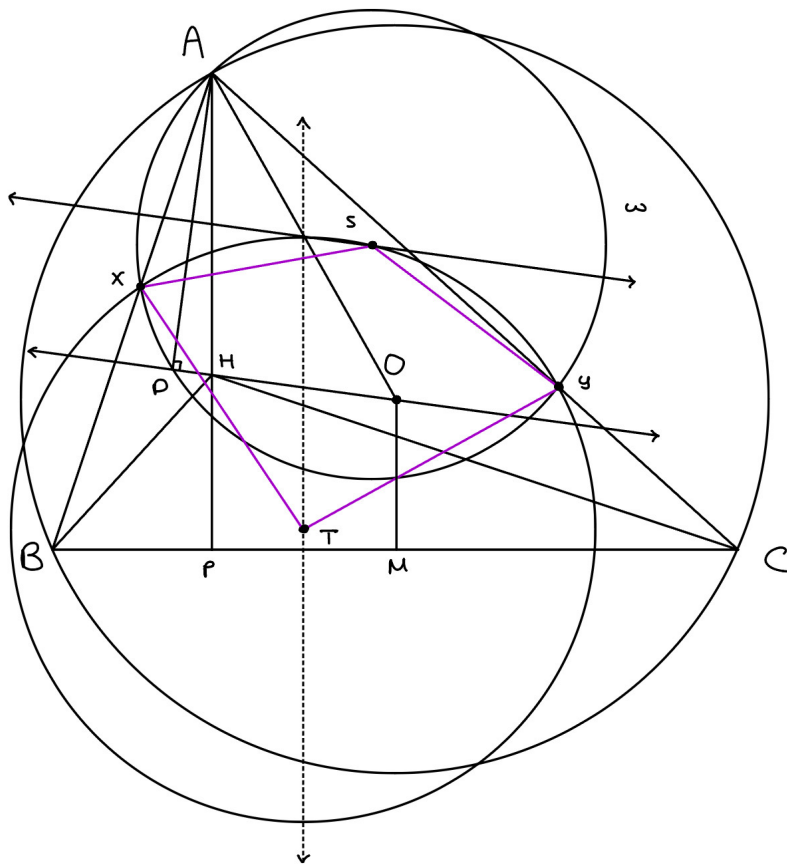
To find this fixed point, let the perpendicular bisector of BC intersect AB and AC at X and Y , respectively. Then by setting $D = B$ and $D = C$ we find that the fixed point must lie on (AXC) and (AYB) . By angle chasing we may verify that this point is the circumcenter O of ABC .



Example 3.7 (ISL 2016 G5) Let D be the foot of perpendicular from A to the Euler line (the line passing through the circumcentre and the orthocentre) of an acute scalene triangle ABC . A circle ω with centre S passes through A and D , and it intersects sides AB and AC at X and Y respectively. Let P be the foot of altitude from A to BC , and let M be the midpoint of BC . Prove that the circumcentre of triangle $XS Y$ is equidistant from P and M .

Linearly move X along AB . Since (AXY) always passes through D , we have that Y also moves linearly. Since $\angle XSY = 2\angle A$ is fixed as X varies, and $SX = SY$ always, we have that as X varies, $XS Y$ is always directly similar to a fixed triangle, and hence S also moves linearly. Now let T denote the circumcenter of $XS Y$. Since $\angle XTY = 360^\circ - 2\angle XSY$ and $TX = TY$, XTY is also always directly similar to a fixed triangle, and hence moves along a fixed line. So it suffices to check that T is equidistant from P and M for two choices of T .

Now, let H be the orthocenter of ABC and O its circumcenter. For one case, consider when X is the foot of the C -altitude. Then ω is the circumcircle of diameter AH , so S is the midpoint of AH , and Y is the foot of the B -altitude. Hence T is the nine-point center, which is equidistant from P and M (as the nine-point circle passes through both). For our second case, consider when X is the midpoint of AB . Then Y is the midpoint of AC , ω is the circle with diameter AO , and S is the midpoint of AO . In this case we have that T lies on the perpendicular bisector of XY ; but the perpendicular bisector of the line connecting the midpoints of AB and AC is precisely the perpendicular bisector of PM , so we are done.



Example 3.8 (ISL 2012 G6) Let ABC be a triangle with circumcenter O and incenter I . The points D, E and F on the sides BC, CA and AB respectively are such that $BD + BF = CA$ and $CD + CE = AB$. The circumcircles of the triangles BFD and CDE intersect at $P \neq D$. Prove that $OP = OI$.

Consider when D is the A -extouch point A_1 . Then we may check that E is the B -extouch point B_1 and F is the C -extouch point C_1 in this case. If K is the reflection of I across O (the so-called *Bevan point*), then we may check that $A_1 = (BK) \cap (CK)$ and so on. So $P = K$ in this case, and we have that O is the midpoint of KI .

Now vary D, E, F . The given condition says that $AE + AF$ is fixed, so (AEF) passes through a fixed point; call it A_0 . Then we know by linearity results that $A_0E = A_0F$. Hence in fact the fixed point lies on AI . In fact, by investigating the case where $D = A_1, E = B_1, F = C_1$, we have that A_0 lies on (AK) , so A_0 is the projection of K onto AI . Hence A_0, B_0, C_0 all lie on (KI) .

To finish, now note that

$$\angle A_0PC_0 = \angle A_0PE + \angle EPC_0 = \frac{1}{2}\angle A + \frac{1}{2}\angle C = 180^\circ - \angle AIC = \angle A_0KC_0.$$

Thus P also lies on (IK) , and hence $OP = OI$ as desired.

